CHAPTER 11

Spectroscopy

Independent exercise solving

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

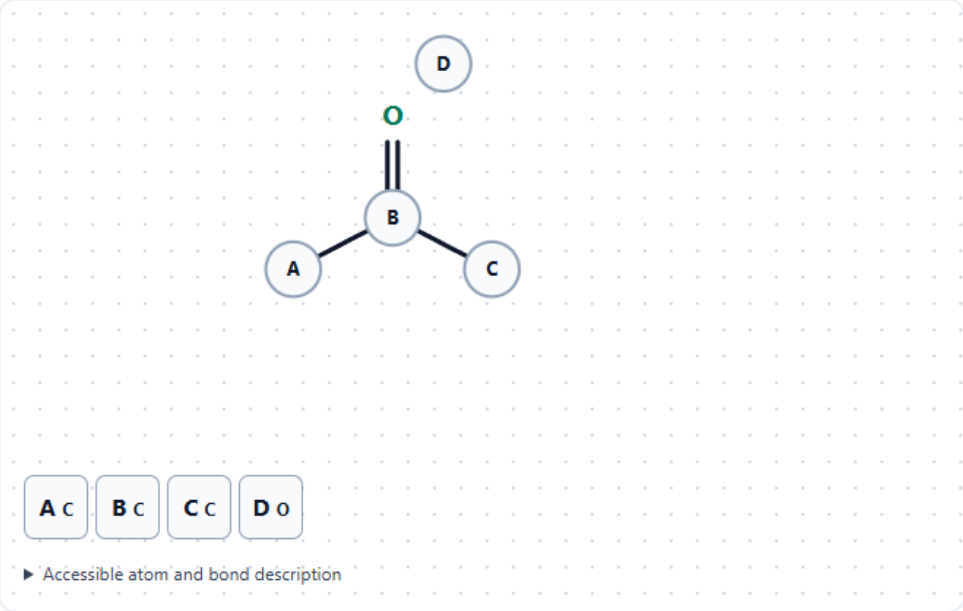
Source snapshot SHA-256: ed978f297b8cf2ece50285a78a7fad18a23424167fb9b8d2941a7e662d626fe7

BLIND PASS: answer keys, author molecule names/formulas, hints, explanations and answer overlays are withheld. Solve every numbered stage independently. Report the stage ID, answer, derivation, and uncertainty. Question wording and displayed diagram captions are retained because they are visible task data.

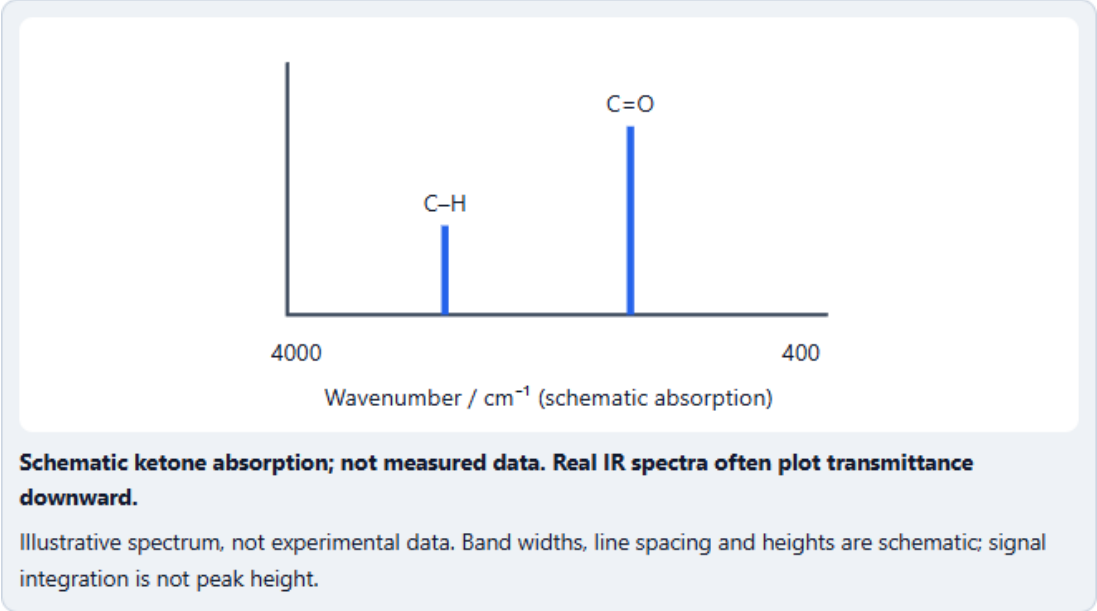
Do not open the full review packet in this chat before saving your independent attempt. The full packet contains keys and lessons that would contaminate a blind attempt.

ir-read | Independent exercise

Screenshot: ir-read-stage-1-molecule



Screenshot: ir-read-diagram



Schematic ketone absorption; not measured data. Real IR spectra often plot transmittance downward.

Illustrative spectrum, not experimental data. Band widths, line spacing and heights are schematic; signal integration is not peak height.

{"diagramSpecification": {"type": "spectrum", "min": 400, "max": 4000, "unit": "Wavenumber / cm^{-1} (schematic absorption)", "caption": "Schematic ketone absorption; not measured data. Real IR spectra often plot transmittance downward.", "peaks": [{"x": 1715, "height": 0.95, "label": "C=O"}, {"x": 2950, "height": 0.45, "label": "C–H"}]}}

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=108.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 2]]

ir-read:stage-1 | Select the bond endpoints responsible for the prominent carbonyl band.

Answer type: atoms

ir-read:stage-2 | An IR-active vibration must change...

Answer type: choice

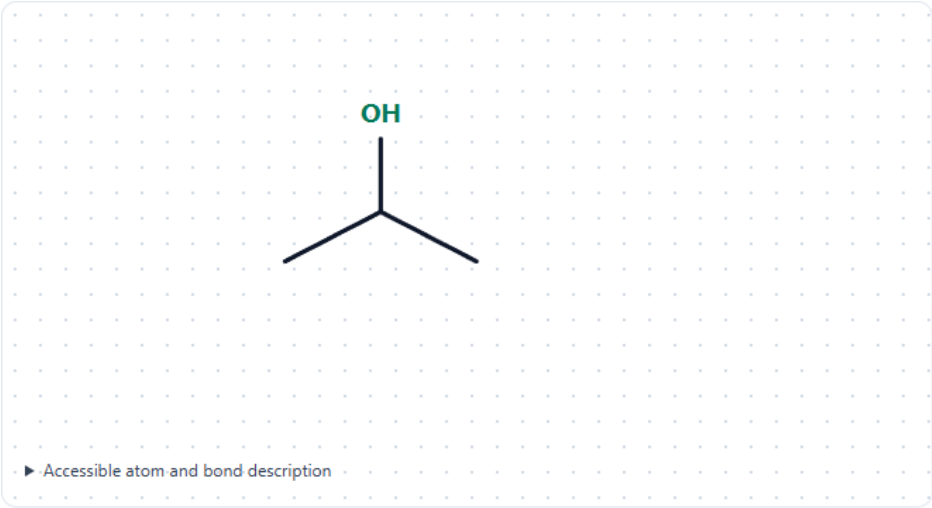
Option 0 (A): Dipole moment

Option 1 (B): Molecular formula

Option 2 (C): The number of carbon atoms

This stage uses a different molecule (connectivity shown).

Screenshot: ir-read-stage-3-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=108.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 1]]

ir-read:stage-3 | A broad 3200–3600 cm^{-1} band appearing after reduction supports...

Answer type: choice

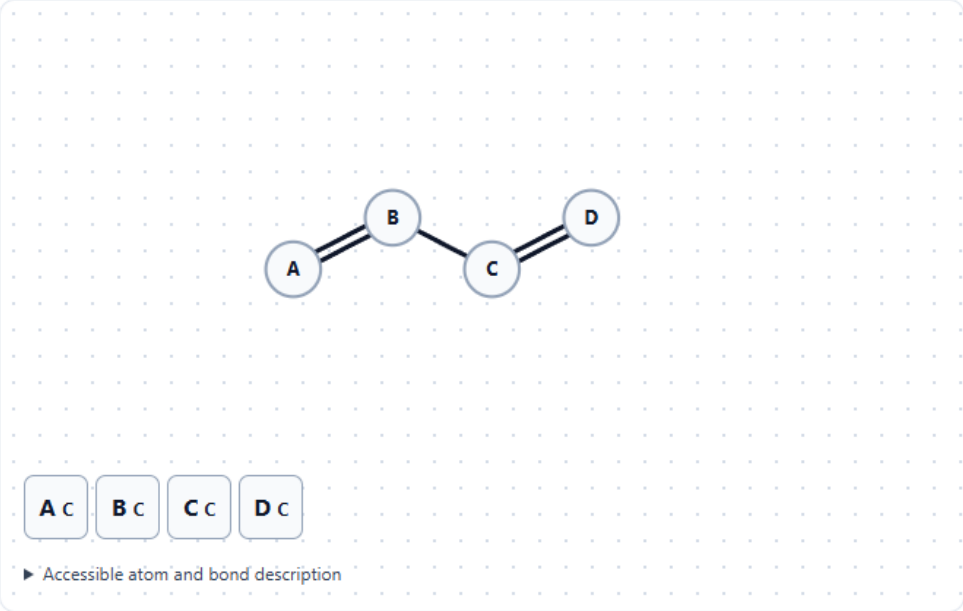
Option 0 (A): An alcohol OH

Option 1 (B): Only a new alkane

Option 2 (C): A nitrile as the sole explanation

uv-energy | Independent exercise

Screenshot: uv-energy-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=224.00, y=85.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2], [1, 2, 1], [2, 3, 2]]

uv-energy:stage-1 | Select all carbons participating in the conjugated diene.

Answer type: atoms

uv-energy:stage-2 | For comparable pi systems, increased conjugation often shifts absorption to…

Answer type: choice

Option 0 (A): Longer wavelength

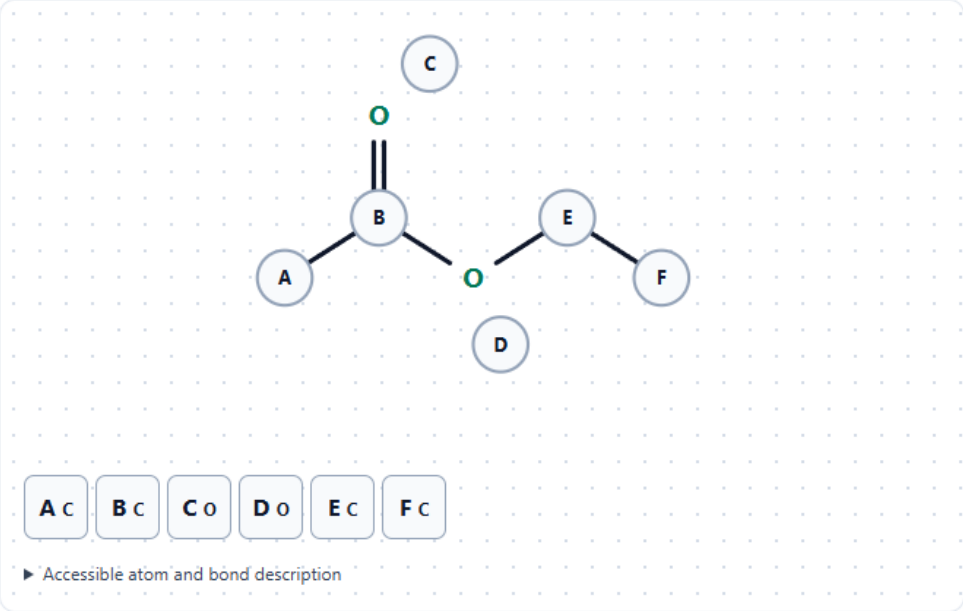
Option 1 (B): Shorter wavelength in every case

uv-energy:stage-3 | If ε=10000 L mol^{-1} cm^{-1}, l=1 cm, and c=0.000020 mol L^{-1}, what is A?

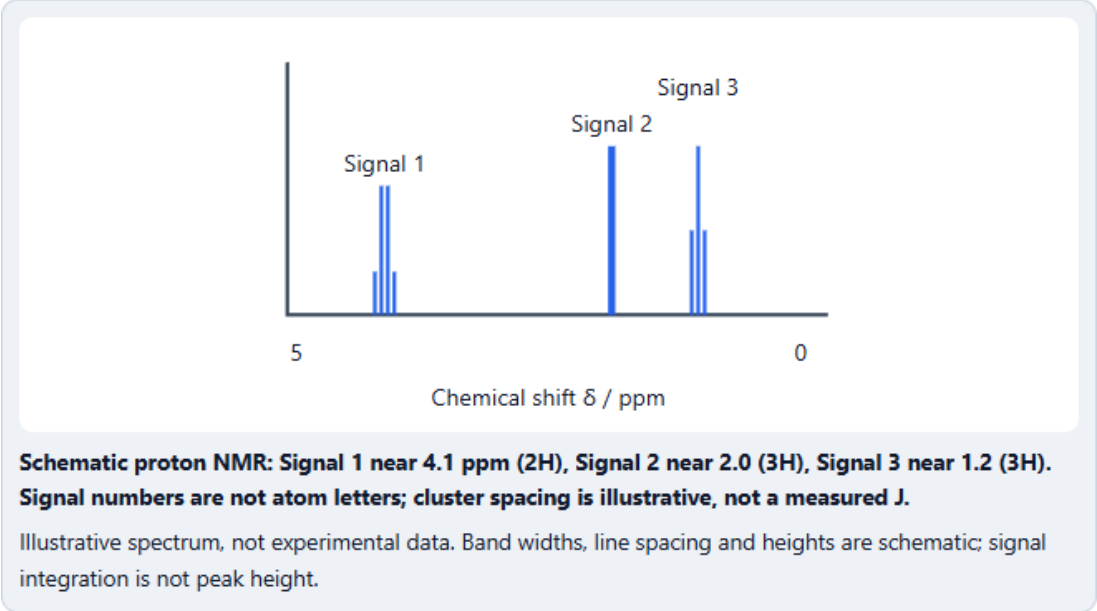
Answer type: number | unit: dimensionless

nmr-ethyl | Independent exercise

Screenshot: nmr-ethyl-stage-1-molecule



Screenshot: nmr-ethyl-diagram



Schematic proton NMR: Signal 1 near 4.1 ppm (2H), Signal 2 near 2.0 (3H), Signal 3 near 1.2 (3H). Signal numbers are not atom letters; cluster spacing is illustrative, not a measured J.

Illustrative spectrum, not experimental data. Band widths, line spacing and heights are schematic; signal integration is not peak height.

{"diagramSpecification": {"type": "spectrum", "min": 0, "max": 5, "unit": "Chemical shift δ / ppm", "caption": "Schematic proton NMR: Signal 1 near 4.1 ppm (2H), Signal 2 near 2.0 (3H), Signal 3 near 1.2 (3H). Signal numbers are not atom letters; cluster spacing is illustrative, not a measured J.", " peaks": [{"x": 4.1, "height": 0.65, "label": "Signal 1", "lines": 4, "spacing": 0.06}, {"x": 2, "height": 0.85, "label": "Signal 2", "lines": 1}, {"x": 1.2, "height": 0.85, "label": "Signal 3", "lines": 3, "spacing": 0.06}]}}

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=45.00, y=120.00; charge=0; explicit H override=none

1 (B): C; x=100.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=100.00, y=25.00; charge=0; explicit H override=none

3 (D): O; x=155.00, y=120.00; charge=0; explicit H override=none

4 (E): C; x=210.00, y=85.00; charge=0; explicit H override=none

5 (F): C; x=265.00, y=120.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [1, 3, 1], [3, 4, 1], [4, 5, 1]]

nmr-ethyl:stage-1 | Select the OCH2 carbon.

Answer type: atoms

nmr-ethyl:stage-2 | How many lines does its signal have in the simple n+1 model?

Answer type: number | unit: lines

nmr-ethyl:stage-3 | The terminal ethyl CH3 gives an approximate…

Answer type: choice

Option 0 (A): Triplet, integrating to 3H

Option 1 (B): Quartet, integrating to 2H

Option 2 (C): Singlet, integrating to 1H

combined-evidence | Independent exercise

Supplied illustrative data: C3H6O; a ketone-like C=O band; one proton NMR singlet integrating to 6H near 2 ppm.

Screenshot: combined-evidence-stage-1-molecule

Candidate structure withheld. Use the supplied observations to infer it; the structure is shown after completion.

combined-evidence:stage-1 | For formula C3H6O, calculate DBE.

Answer type: number | unit: DBE

combined-evidence:stage-2 | A single 6H singlet near 2 ppm suggests...

Answer type: choice

Option 0 (A): Two equivalent methyl groups with no adjacent H-bearing carbon

Option 1 (B): Six nonequivalent one-H environments

Option 2 (C): An aldehydic proton at 2 ppm

combined-evidence:stage-3 | With a ketone-like IR band, name the consistent C3H6O structure.

Answer type: text